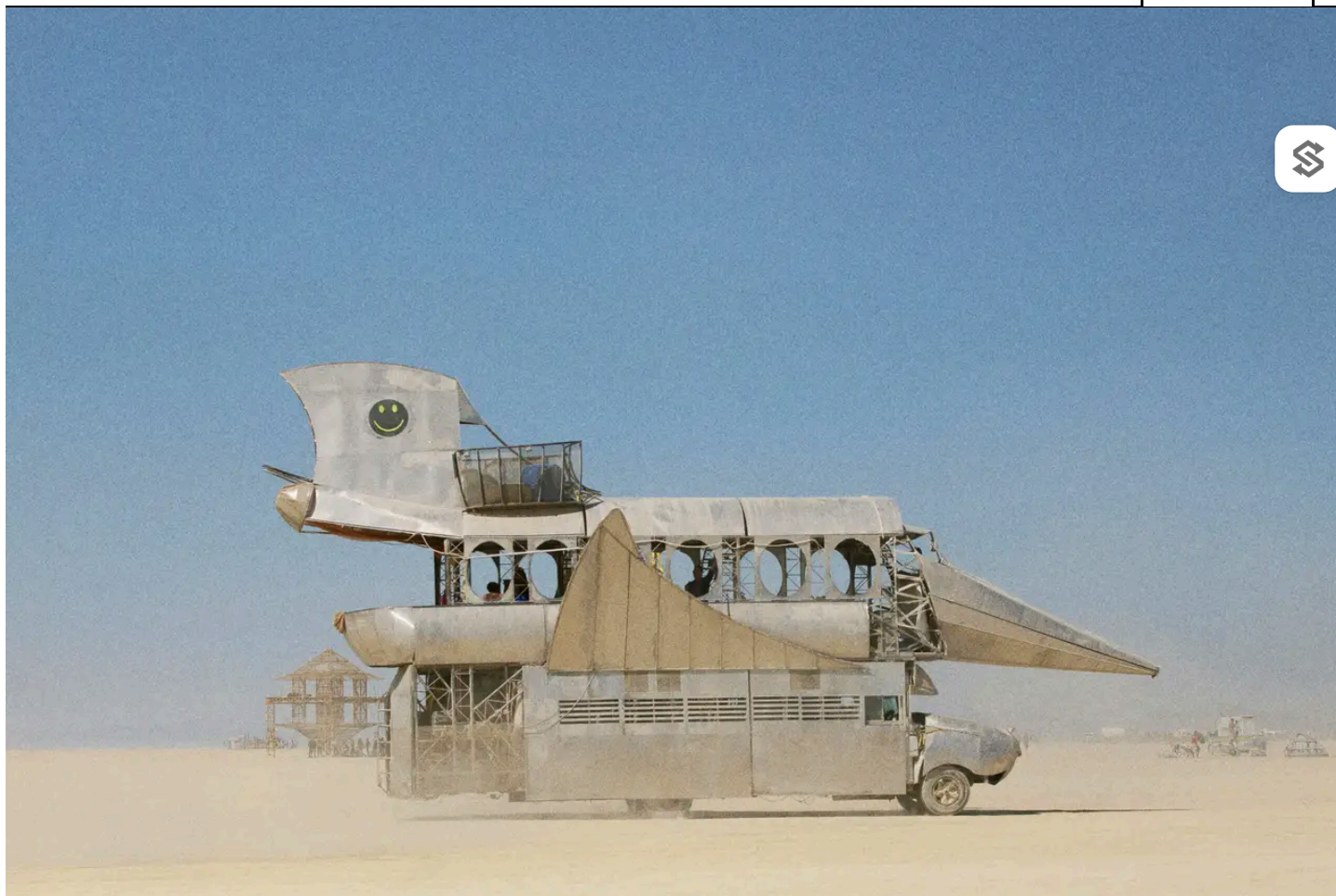




Слова , произнесенные Мириам Пински

Всеобъемлющая теория общественного транспорта

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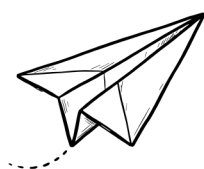
8 Минут

Кому выгодно, когда мы просим транспортную отрасль преследовать социальные цели?

Мы требуем от общественного транспорта решения проблем, для которых он никогда не предназначался. Велосипедные сервисы должны сокращать выбросы и улучшать качество воздуха в городах. Станции метро должны украшать окрестности. Автобусы нужны для создания рабочих мест. И каждый новый транспортный проект должен учитывать мнение местных жителей и расширять их возможности. Все это мешает общественному транспорту выполнять свою главную задачу — перевозить людей.



Все эти цели имеют право на существование. Но каждая из них требует денег и приводит к задержкам. В проекте «Транзитные издержки» рассказывается о том, как на участке бостонской «Зелёной линии», строительство которого завершилось на десять лет позже, чем планировалось, появилась трёхкилометровая велосипедная и пешеходная дорожка после того, как 13 человек выступили за её строительство на собрании сообщества. Стоимость проекта составила ошеломляющие 20 миллионов долларов.



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Любая второстепенная цель вступает в противоречие с необходимостью перевозить как можно больше людей и с благополучием малоимущих, которые чаще всего пользуются общественным транспортом. Хотя 91,5 процента американских семей владеют хотя бы одним автомобилем, менее половины (45 процентов) пассажиров общественного транспорта имеют возможность пользоваться личным автомобилем дома.

Доля автобусов в общем количестве транспортных средств на дорогах ничтожно мала: на их долю приходится менее одного процента выбросов в США. Тем не менее на них распространяется больше правил и норм, чем на личные автомобили. Правительства строят дороги, автомагистрали и парковки, уделяя гораздо больше внимания своей основной задаче — повышению мобильности. Однако транспортные компании подчиняются гораздо большему количеству требований.



Побочные задания на транзит

Транзитные перевозки не всегда были медленными. До середины XX века почти все виды общественного транспорта находились в частной собственности, и цели были простыми: максимизировать пассажиропоток и доходы от продажи билетов при минимальных затратах. Но начиная с 1910-х годов транспортные компании столкнулись с конкуренцией со стороны автомобилей.

В 1960-х годах правительства начали брать на себя управление пришедшими в упадок транспортными линиями. Это привело к возникновению новой проблемы: общественный транспорт стал зависеть от субсидий налогоплательщиков, при этом большая часть финансирования поступала от избирателей, которые им не пользовались. Большинство людей перешло на личный транспорт, поэтому необходимость в субсидиях приходилось обосновывать интересами тех, кто ими не пользовался. Общественный транспорт стал частью растущих социальных движений 1960-х и 1970-х годов: войны с бедностью, движения за гражданские права, защиты окружающей среды и борьбы с нефтяным кризисом. Сегодня Американская ассоциация общественного транспорта преподносит общественный транспорт как панацею: инвестиции в общественный транспорт, по ее словам, «помогут американцам добираться до работы, учебных заведений и медицинских учреждений, а также будут способствовать созданию новых рабочих мест, уменьшению пробок на дорогах и улучшению качества воздуха».

The typical transit agency gets funding from federal, state, and local sources. Each level of government can add more goals. The federal government might require domestically sourced materials (such as Buy America), daycare for manufacturers (like the CHIPS Act), or a higher minimum wage for contractors

and subcontractors (as is the case for Davis-Bacon prevailing wage rules). States might add extra environmental review (like the California Environmental Quality Act). And local governments can shape design, public art elements, local hiring, and the means of conducting community engagement.

The transit tradeoff

Alongside my co-authors, I studied six transportation pilot programs that were tasked with various goals, among them: reducing emissions, improving mobility equity, and reducing miles driven in cars. All of the pilots, which included electric carshare, bikeshare, and microtransit services, were funded by the California Air Resources Board and were located in disadvantaged areas across California.

By and large, these programs did improve the mobility of the badly off: most users lived in households earning under \$25,000 a year, and many of the trips were necessary to get to work, school, or medical appointments. But the programs added 128,880 net new vehicle miles and slightly increased overall emissions rather than decreasing them. As mobility schemes, they worked, but on other margins they did not.

The problem is not just specific equity or environmental programs. Existing services are being eroded by attempts to make buses electric. Battery electric buses cost more to procure, and parts are more expensive to replace. Austin's CapMetro 40-foot electric buses cost \$917,000 apiece, whereas their diesel buses cost just \$535,000. In south Florida, the Miami-Dade county's transit agency spent approximately \$1.1 million per electric bus. On any given day, out of the 75 buses it purchased, only about six are operating. That's because the electric buses break down every 600 miles (seven times as often as their diesel equivalents), and because parts are difficult to come by, particularly since the manufacturer, Proterra, filed for bankruptcy.

In Germany, electrifying bus fleets increased costs by 12 percent. The problem is even worse in the United States because of Buy American rules. In Europe, where Chinese manufacturers compete freely, a standard 12-meter electric bus sells for roughly \$600,000. US agencies pay nearly twice as much. While demand for electric buses is growing, the US market remains small, leaving agencies with few competitors to choose from. Out of roughly 63,000 buses in the US, approximately 7,000 are electric, 5,000 of which are school buses.

Moving from diesel to electric buses doesn't just mean swapping vehicles: it also requires new software, land use permitting, grid connections for charging infrastructure, and worker time to fuel vehicles. Just routing electric buses around charging infrastructure adds significant complexity to operating the service.

Every dollar spent complying with Buy America or staging community meetings is a dollar not spent improving frequency or coverage. Assuming the median cost per hour for operating a bus in the US is \$131, and that the median US bus carries 14.3 passengers per hour, the money for electrifying one bus could fund roughly 2,300 additional service hours, carrying at least 30,000 additional riders.



Prices make perfect

Electrifying transit is one way to reduce emissions, but it's not the most efficient. Researchers studying Stockholm's busiest bus route found that better pricing roads and transit, and running buses more often, yielded more benefit than just electrifying buses. From a passenger perspective, electric buses are no different. But operators lose out because costs rise by at least 16 percent, depending on the charging setup. Combining road charges with more frequent buses entices drivers to shift to transit, reducing emissions more substantially than electrifying a few buses.

When transit becomes more viable and reliable, we see a positive feedback loop. People who would otherwise drive opt for transit, which boosts revenue and feeds back into improved quality. On average, increasing transit service frequency by 1 percent boosts ridership by about 0.5 percent. Once buses arrive every ten minutes, bus services become more appealing for riders with alternatives. These switches are the only kind that can make a substantial dent in emissions.

Congestion pricing would reduce emissions much more effectively than electrifying buses. In the few places brave enough to try, the outcomes have been significant. In London, greenhouse gas emissions from vehicles fell by 16 percent, and the charge generated £227 million in a year. As well as this, traffic accidents fell almost immediately, with nine percent fewer accidents in the congestion charge's first year.

It's much easier, of course, to encourage transit ridership in places where people and destinations cluster together. Switching people from driving to transit saved nine million metric tons of carbon in 2018. But we can prevent 66 million metric tons of carbon emissions by decreasing the distances people need to travel through more efficient land use. When distances are short, walking, biking, and transit all become more appealing, and when people do drive, they drive shorter distances. In major cities, where transit is more dominant, electrifying buses makes more sense than it does in car-dominant places, like Fresno, California, where it's more important to move low-income people to their jobs. 

Focusing transit on what it does best

In most of the US, not having a car makes poverty worse. Poor people who gain access to a car are more likely to find jobs, keep their jobs, find higher paying jobs, work more hours, get off welfare, and move to better neighborhoods. Even beyond economic outcomes, people who don't have cars take part in fewer activities outside the home, have fewer extracurricular activities and schools to choose from, fewer grocery store options, and are more likely to delay medical care.

Scope creep fails on its own terms. We are asking people already struggling to make ends meet, who already emit far less than most Americans do, to save the planet. But this is self defeating. Every dollar we spend on the environment or union jobs is a dollar we don't spend on making transit more competitive against cars.

A 2018 UCLA report on transit ridership in Southern California estimated that if just one in every four people who rarely rides transit replaced a single driving trip with transit every two weeks, annual ridership would grow by 96 million. It's more sensible to reduce emissions by getting a broader base to make small behavioral changes than to force a small group of already burdened people to carry the load.



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